Alternative Assessment Tool(AAT) Report on

"Simulation of sequential and combinational circuit in cadence tool "

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT-2) for the course

ADVANCED DIGITAL LOGIC DESIGN

[Course code: 23EC5PE1AD]

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github.com/amrutha-ck28/ADLD_AAT

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Experiment 1(Combinational Circuit) -4bit Comparator

Code:

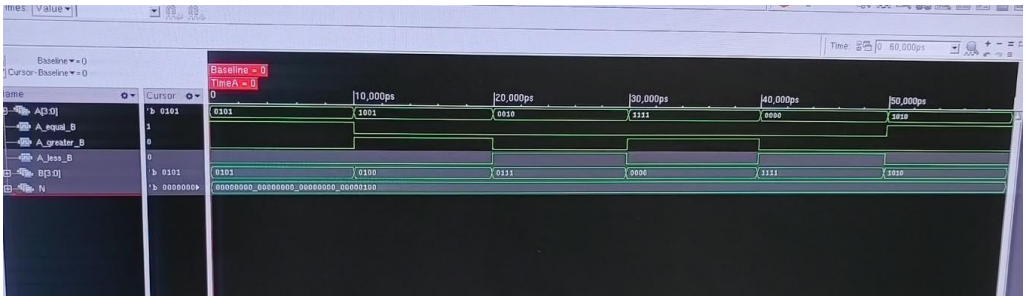
```
module comp4(
   input  [3:0] A,
   input  [3:0] B,
   output A_equal_B,
   output A_greater_B,
   output A_less_B
);
assign A_equal_B   = (A == B);
assign A_greater_B = (A > B);
assign A_less_B    = (A < B);
endmodule
```

Testbench:

```
`timescale 1ns/1ps
module tb_comp4;
reg  [3:0] A, B;
wire A_equal_B;
wire A_greater_B;
wire A_less_B;
comp4 uut(A,B ,A_equal_B, A_greater_B, A_less_B);
initialbegin
   A = 4'b0101; B = 4'b0101;#10;
   A = 4'b1001; B = 4'b0100;#10;
   A = 4'b0010; B = 4'b0111; #10;
   A = 4'b1111; B = 4'b0000; #10;
```

```
    A = 4'b0000; B = 4'b1111;#10;
    A = 4'b1010; B = 4'b1010; #10;
    $finish;
end
initial
begin
    $monitor("Time=%0t A=%b B=%b EQ=%b GT=%b LT=%b",
            $time,A,B,A_equal_B,A_greater_B,A_less_B);
end
endmodule
```

Simulation Results:



Sequential Circuit : Shift Regiter

Code:

```
`timescale 1ns/1ps
module shift_register #(parameter N=8)(
    input wire clk,
    input wire rst,
    input wire shift_en,
    input wire d_in,
```

```
    output reg [N-1:0] q_out
);
always @(posedge clk) begin
    if (rst)
        q_out <= {N{1'b0}};
    else if (shift_en)
        q_out <= {q_out[N-2:0], d_in};
end
endmodule
```

Testbench:

```
`timescale 1ns/1ps
module tb_shift_register;
parameter N=8;
reg clk,rst,shift_en,d_in;
wire [N-1:0] q_out;
shift_register #(N) uut(.clk(clk),.rst(rst),.shift_en(shift_en),.d_in(d_in),.q_out(q_out));
always #5 clk=~clk;
initial begin
    $dumpfile("shift_register_waves.vcd");
    $dumpvars(0,tb_shift_register);
    clk=0; rst=1; shift_en=0; d_in=0;
    #10 rst=0; shift_en=1; d_in=1;
    #10 d_in=0;
    #10 d_in=1;
    #10 d_in=1;
    #50 shift_en=0; d_in=0;
    #20 $finish;
end
```

endmodule

Simulation result:

